

1. Introduction and Objective

The aim of this project was to develop and evaluate two baseline machine learning models for predicting Emergency Severity Index (ESI) levels from emergency department triage data. The purpose of creating these baseline models was not to build a final clinical decision system, but to establish whether simple and interpretable machine learning approaches could identify patterns associated with patient urgency. The two models developed were Logistic Regression and Decision Tree classifiers. These models were compared against a stratified random baseline to determine whether they provided meaningful improvement over simple guessing. Because emergency triage is a safety-critical setting, model performance was evaluated using clinically relevant measures rather than accuracy alone. The main clinical concern is correctly identifying the most urgent patients, particularly those classified as ESI Level 1. Missing a critically ill patient could delay emergency intervention, making the consequences of some errors much more serious than others.

2. Dataset Preparation and Model Development

The models were trained using the cleaned Yale EMLLC emergency department triage dataset. The target variable was the patient's ESI level, where ESI Level 1 represents the highest urgency and ESI Level 5 represents the lowest urgency. The dataset was divided into training and testing sets using an 80/20 split. Stratification was applied to preserve the distribution of ESI levels across both sets. A fixed random seed of 42 was used to ensure that the results were reproducible. The Logistic Regression model required feature scaling because it is affected by differences in feature ranges. A StandardScaler was fitted only on the training data and then applied to the test data. The Decision Tree model was trained without scaling because tree-based models make decisions through feature thresholds and are not affected by differences in numerical ranges. The tree depth was limited using `max_depth=5`. This was chosen to reduce overfitting and maintain interpretability, allowing the decision-making process to remain easier to understand in a clinical setting. Both models were compared against a Dummy Baseline using a stratified random prediction strategy. This provided a baseline representing performance from a model that does not learn meaningful patterns from the data.

3. Model Results

The baseline comparison showed that both machine learning models performed better than the random baseline.

Table 1. Baseline Model Performance Comparison

	Model	Accuracy	Macro F1	ESI-1 Recall
0	Dummy Baseline	0.375	0.204	0.00
1	Logistic Regression	0.667	0.495	0.25
2	Decision Tree	0.556	0.216	0.00

Logistic Regression achieved the strongest overall performance, with the highest accuracy and macro F1 score. It also performed better at identifying ESI Level 1 patients, although its recall remained limited. The Decision Tree provided a more interpretable structure but struggled to identify the rarest and most critical class. While the difference between macro F1 and weighted F1 highlights the impact of class imbalance in this dataset. Weighted F1 gives more influence to common classes, meaning strong performance on frequent ESI levels can increase the score. Macro F1 treats every ESI class equally, ensuring that rare but clinically important classes are included in the evaluation. For triage applications, macro-level evaluation is important because performance on minority classes cannot be ignored.

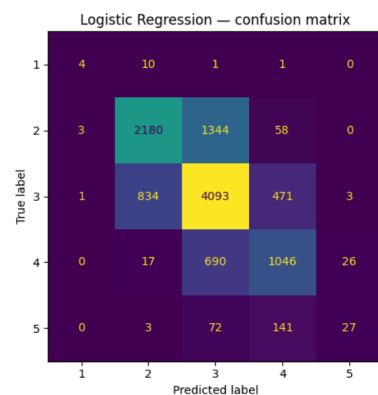


Figure 1: Logistic Regression Confusion Matrix

The confusion matrix demonstrates where the Logistic Regression model correctly classified patients and where errors occurred between ESI categories.

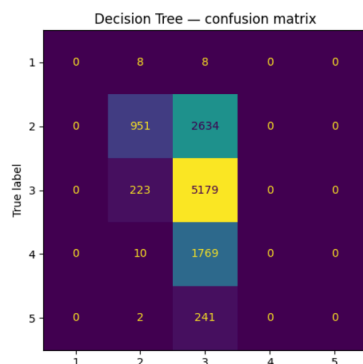


Figure 2. Decision Tree Confusion Matrix

The Decision Tree confusion matrix shows that the model frequently predicted common ESI categories while struggling with rare classes, particularly ESI Level 1.

4. Primary Metric Justification

The primary metric selected for this project is recall for ESI Level 1. In emergency medicine, the most important concern is identifying patients who require immediate intervention. ESI Level 1 patients represent the highest-risk group, and failing to recognise these patients could result in delayed treatment, worsening clinical outcomes, or avoidable harm. Overall accuracy is not sufficient for this problem because the dataset contains many more lower-acuity patients than critically ill patients. A model could achieve a high accuracy score by correctly predicting common lower-risk cases while still missing the small number of patients who need urgent care. Measuring how many true ESI Level 1 patients are successfully identified provides a more clinically meaningful assessment of whether the model is safe enough to support triage decisions.

5. Failure Mode Reflection

The main failure mode identified was the inability of the models to consistently detect ESI Level 1 patients. The Decision Tree model achieved an ESI Level 1 recall of 0, meaning it failed to identify any patients in this highest urgency category within the test set. Logistic Regression performed better, identifying some ESI Level 1 cases with a recall of 0.25, but it still missed many critically ill patients. In a real emergency department environment, these missed cases represent the greatest safety concern. A patient requiring immediate resuscitation could potentially be incorrectly classified as a lower urgency level, causing delays before receiving necessary care. Future model development should focus on improving detection of rare high-risk cases while maintaining transparency and clinical safety.

6. Conclusion

This project successfully developed two baseline triage prediction models using Logistic Regression and Decision Tree classifiers. Both models improved upon the random baseline, demonstrating that machine learning can identify patterns within emergency department triage data. However, the evaluation also showed important limitations. The models performed less effectively on rare but clinically significant ESI Level 1 cases. For future development, improving sensitivity to critical patients should be prioritised over simply increasing overall accuracy. These baseline models provide a foundation for further investigation while highlighting the importance of clinically appropriate evaluation methods.